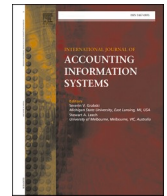




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Challenges and opportunities for artificial intelligence in auditing: Evidence from the field[☆]

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ABSTRACT

In this study we research the adoption of artificial intelligence (AI) in auditing by large public accounting firms, with emphasis on its challenges and opportunities. Some previous studies point to delayed adoption of AI in auditing due to regulations and the need for additional safeguards while others document extensive AI implementation. To address this dissensus, we conducted 22 interviews with experienced audit professionals. We find that “simple AI” technologies such as key data extraction from documents and optical character recognition are used widely in audits while “complex AI” tools are only being developed. We find RPA is used to automate repetitive administrative processes while the use of RPA for audit tasks is not as common. We also find that the main AI adoption challenges are related to transparency and explainability, AI bias, data privacy, robustness and reliability, fear of auditor overreliance on AI, and the need for AI guidance. We present ideas for addressing these challenges based on our research and lessons from other fields.

1. Introduction

This paper is motivated by the need to develop artificial intelligence (AI) governance mechanisms for trustworthy AI development and use in audits.¹ Before these complex mechanisms can be developed, however, it is first necessary to gain some clarity on the risks and opportunities AI introduces to audit firms as they develop and use AI-aided audit support systems and their clients whose AI use influences their risk profile and thus the audit approach and the tools deployed as part of the audit. Therefore, we research (a) how AI is being implemented in auditing, (b) how audit clients are using AI in financial reporting, (c) what are the challenges related to AI implementation, and (d) what are the solutions to these challenges proposed by auditors and found in other fields.

Our work explores how AI is being used in audits, with special emphasis on the challenges and potential solutions related to the adoption of AI in audit. We provide illustrative examples based on interviews with 22 experienced audit professionals. Several studies

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¹ We define AI “as a set of technologies that use such capabilities as knowledge, insight, and perception to perform tasks that only humans could perform previously” (Davenport 2018).

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call for more research in this area (Issa et al 2016; Kokina and Davenport 2017; Almufadda and Almezeini 2022). Boritz and Stratopoulos (2023) call for “AIS research aimed at helping us understand the strengths and weaknesses of AI technologies, the benefits, costs, and risks that they bring” and urge the AIS community to research these issues “ideally in collaboration with practitioners in industry and public practice” (p.7). In addition, Fotoh and Lorentzon (2021) encourage more research to address “dilemmas that audit firms face in applying digital audit procedures” (p. 91).

Our paper makes a necessary contribution to the growing literature on AI adoption in auditing. We provide a systematic overview and categorization of AI implementation challenges, including those that are linked to the human elements as well as technological aspects of AI and the nature of the audit process itself. Recognizing that the interface between technology and human judgement can be a source of challenges (e.g., Munoko et al 2020; Samiolo et al 2023), our research confirms that these issues are serious considerations for audit professionals, and uncovers additional roadblocks to AI adoption in audit as well as dilemmas and uncertainties encountered by firms as they implement AI tools. While current literature mostly focuses on documenting some aspects of issues related to AI adoption in auditing (Samiolo et al 2023; Boritz and Stratopoulos 2023; Seethamraju and Hecimovic 2022; Lehner, Ittonen, Silvola, Strom, and Wuhrlleitner 2022; Fotoh and Lorentzon 2022; Fedyk, Hodson, Khimich, and Fedyk 2022; Munoko, Brown-Liburd, and Vasarhelyi 2020), studies addressing approaches to manage these challenges are scarce (e.g., Zhang, Cho, and Vasarhelyi, 2022). Importantly, a unique contribution of our paper is that we study potential solutions to these challenges through our interviews as well as through developments in fields other than auditing.

Through our interviews with experienced audit professionals, we find that large firms are adopting various AI technologies. However, an analysis of AI examples provided by the participants reveals a more nuanced understanding of AI implementation. We find that “simple AI” technologies (e.g., simple machine learning, computational linguistics) are used widely in audits while “complex AI” tools (e.g., deep learning neural networks, generative AI) are only being adopted experimentally but are not yet fully deployed into production processes. However, the more advanced AI technologies are maturing rapidly and are likely to become institutionalized. Similar to Fedyk et al (2022), we find that the firms are using RPA to automate repetitive administrative processes while the use of RPA for audit tasks is not as common. While some observers do not view RPA as AI, it does take automated digital actions, makes decisions with a rule engine, and is increasingly combined with machine learning.

Specifically, we find that auditors use simple natural language processing tools to extract key information from contract documents. They are using machine learning to detect anomalies in financial information and prioritize further human investigation. They are also investing heavily in data platforms to make further use of AI feasible. This AI activity has not yet diminished the need for human auditors, but it may do so in the future. AI typically automates tasks rather than entire jobs, but a large degree of audit task automation might lead to a need for fewer human auditors, e.g., those at the entry level. As with other jobs, however, we believe that the most common outcome for auditors will be working alongside AI rather than being replaced by it (Davenport and Miller 2022).

We find that the main challenges in AI tool development are related to transparency and explainability, AI bias, data privacy, robustness and reliability, fear of auditor overreliance on AI, and the need for AI guidance. Study participants report that audit clients are not yet using complex AI in financial reporting, but that the interaction between AI-aided financial reporting by clients and auditing could become a novel challenge in the future. While new techniques to audit client AI are being developed, the lack of trust in AI tools discourages auditors from auditing through AI and instead they audit around it by reviewing the outputs the tools generate.

The findings of this study should be of interest to academics exploring emerging technology adoption in auditing and auditors in practice trying to learn from professionals in their own field and other fields. Understanding the challenges and opportunities associated with AI is important for professional service firms as they assess ways to balance audit quality and efficiency. In addition, implementing responsible AI will not only be important to the firms but to their clients. Company managers can benefit from learning about the tools that auditors will implement to audit their financial statements and evaluating the opportunities and risks associated with their client service provider using AI tools with their company data. Continuing to provide additional insight into how public accounting firms are using technology can also inform the PCAOB and other regulators as they consider changes to the audit standards.

This paper follows the following structure: first, we provide an overview of AI and AI capabilities in accounting and auditing. Then, we present a review of literature on AI adoption in auditing and challenges to AI adoption in auditing. Next, we outline our research method and the context of this study. In the following sections, we provide a detailed analysis of interview results. Then, we discuss the challenges of AI adoption by providing a summary of lessons from our interviews and other fields. In the final section, we provide a summary of our findings, and outline avenues for future research.

2. Background and literature review

2.1. Overview of AI and AI capabilities in accounting and auditing

Artificial intelligence is generally defined as a set of technologies that can perform cognitive tasks that only humans could perform previously (Davenport 2018). The underlying capabilities of AI are statistics, logic (usually in the form of rules), and semantics. These capabilities can be combined or refined into specific applications such as machine learning-based prediction, robotic process automation (RPA) based on rules, and natural language processing based on semantics-based computational linguistics, deep learning neural networks, or both. Combinations of AI technologies are increasingly prevalent in particular business use cases, such as the combination of ML and RPA, and rules engines and ML. Some AI technologies, such as RPA, simple machine learning, key data extraction from documents, and optical character recognition, are easy to use and quite well-established because of the accessibility of predeveloped tools for implementation in audit. We refer to those as “simple AI.” Others, such as deep learning neural networks and generative AI, we refer to as “complex AI” because they are more technically complex and typically require data science and other

technical expertise to create and use effectively in-house. In Table 2, we provide a high level overview of AI technologies and the typical use cases in audit. Technologies that we view under the umbrella of AI are consistent with Boritz and Stratopoulos (2023).

Key technologies for accounting and auditing are machine learning (almost exclusively “supervised” ML, which is trained on data with known outcomes) and automation. ML is the fastest-growing AI technology in business; it is essentially a “prediction machine” (Agrawal, Goldfarb, and Gans 2018) in which a statistical model predicts future values of an outcome variable based on the past data upon which the model has been “trained.” It encompasses traditional machine learning (in its simpler forms, also known as predictive analytics), neural networks, and deep learning neural networks, and generative AI.

Machine learning has many potential (and increasingly actual) uses in accounting and auditing (Haq, Abatemarco, and Hoops 2020). If trained on past accounting data (ideally across organizations), it can predict optimal, normal, or problematic levels of key financial resources—from inventory to accounts payable to risk. Our research suggests that some firms are interested in using anonymized cross-client data to train models, while others are reluctant to do so for client data privacy reasons. ML can make possible continuous auditing of full data sets, rather than samples. ML systems can extract relatively important provisions from contracts and other key documents. They can prioritize particular cases or situations for human investigation. They can integrate data across disparate systems using a “probabilistic matching” approach. Deep learning ML systems can count physical inventory through image recognition.

The other primary AI technology used in accounting is robotic process automation or RPA (Moffitt, Rozario, and Vasarhelyi 2018). RPA is most commonly used for automating structured data-oriented processes, of which there are many in accounting and auditing (Cohen, Rozario, and Zhang 2019). Some uses of RPA and other automation tools include automation of evidence collection, automated report generation, combining data or evidence from multiple sources, and executing tests in Excel or other programs. Samson, Ellis, and Black (2023) document the use of RPA bots in the audits of investment trusts in Australia in instances where APIs (Application Programming Interfaces) were not available to pull data or documents from various sources.

Though basic RPA applications can make some decisions using rules, companies are increasingly combining RPA with machine learning for more sophisticated and data-based decision-making or prediction. This combination, often called intelligent process automation, allows auditors to train models to recognize anomalies or prioritize investigations within a broader automated task flow. Intelligent automation might also include automated generation of reports using generative AI systems, which employ deep learning models to predict the next word in a sequence. Generative AI systems are already being used for a variety of business purposes involving content and code generation (Davenport and Mittal 2022). They produce high-quality drafts of text documents, but need to be prompted, edited, and verified by a human accountant.

2.2. AI adoption in auditing

Recent studies report great variability and little consensus with regard to the levels of AI adoption in auditing. On the one hand, AI adoption in auditing is delayed because the use of emerging technologies is not required by auditing standards (Fotouh and Lorentzon 2022; Issa, Sun, and Vasarhelyi 2016). Another reason for delayed adoption can be attributed to firms wanting additional safeguards in place due to the audit regulatory environment (Cooper, Holderness, Sorensen, and Wood 2019). Cooper et al (2019) conducted 14 interviews with RPA leaders from Big-4 firms to assess the implementation of RPA in public accounting. They report lower levels of RPA development and adoption in auditing compared to taxation and advisory services mainly due to the additional scrutiny associated with the regulation of public company audits. Cooper et al (2019) note that despite the more cautious approach to RPA adoption in audits, the audit practice is particularly suitable for RPA scaling as audits follow a global methodology; taxation, however, relies on a great degree of customization to local legal environments. The authors also note that they expect the reliance on RPA bots in auditing to grow and that at the time the firms were working on using RPA for audit testing. Finally, they report that RPA is viewed as precursor to wider AI adoption in public accounting (Cooper et al 2019). Despite the potentially favorable conditions for RPA use in audits, Eulerich, Pawlowski, Waddoups, and Wood (2022) continue to recognize the slow RPA adoption in audits and high RPA failure rates. To assist auditors in selecting tasks for automation with RPA, they create a three step framework. Fedyk et al. (2022) interview 17 audit partners and national technology leaders of 8 largest firms. Confirming Cooper et al (2019), as it relates to technology adoption in audit, they find that RPA is mostly used for internal administrative tasks and that its use has declined. Bakarich and O'Brien (2021) find that neither RPA nor machine learning (ML) are used widely by public accountants and their clients, yet greater use of these technologies is anticipated in the near future. Samiolo et al (2023) document the common use of data analytics, yet mention that “the actual use of AI we observed was limited to experiments with simple forms of ML” (p. 14). Finally, reflecting on a workshop that brought members of accounting academia and practice, Boritz and Stratopoulos (2023) refer to an influential computer science professor stating that AI use cases mentioned by the panelists “were not really AI but just smarter ways of organizing data” (p. 6).

On the other hand, Issa et al. (2016) predicted that disruptive technological developments in scanning and OCR, cloud, electronic records, and smart contracts would cause auditing to improve its methods and processes. They called for future research to explore the ways in which AI would impact the audit process. Furthermore, Kokina and Davenport (2017) provide examples of technology use through a framework categorizing technologies by task type and level of intelligence. They report that innovation in audit has focused on areas such as client request lists, document review, confirmations, inventory counts, disclosure research, and predictive risk analytics. More recently, Fedyk et al (2022) find that AI use in audits is widespread and state that “we received a strong message that audit firms (Big 4 and non-Big 4) extensively use technology, with AI being one of the most promising and effective instruments to improve quality and efficiency” (p. 945). They report that AI is mainly used to identify anomalies and spot potential fraud with ML for pattern recognition, to match revenue-related transactions such as orders with invoices and receivables with cash receipts, to review contracts and leases with OCR, and to perform benchmarking based on publicly available big data. They document that AI use leads to

improved audit quality and lower audit fees. Other AI-related benefit examples are improvements in areas such as selecting outliers for examination, ability to test the entire population, automated confirmations, three-way testing, reconciliations, analytical procedures, and audit inquiry administration [Seethamraju and Hecimovic \(2022\)](#). Some of these tasks were possible with earlier rule-based technologies, but more complex AI allows for more data-driven and predictive decisions in audit analyses. In sum, while recent papers mostly suggest that there is a slower adoption of AI in audits, it is quite likely that recent rapid developments in AI technology and their ease of use are enabling a faster and wider adoption of various AI tools, especially among larger firms due to the vast investments they have made to advance AI tool development and use.

2.3. Challenges to AI adoption in auditing

[Kokina and Davenport \(2017\)](#) find that one of the main barriers to AI implementation in audit is achieving consistency in data formats across clients. This is confirmed in a literature review by [Almufadda and Almezeini \(2022\)](#), who present a brief summary of examples of AI tools developed by each major accounting firm. They state that incomplete or poor-quality data as well as algorithmic bias and the lack of scalable AI infrastructure are potential risks of using AI. They call for more research, particularly interview-based methods, investigating AI implementation and its risks in audits. Similarly, [Seethamraju and Hecimovic \(2022\)](#) point to technology-related challenges involving the low maturity of AI tools, poor client data quality, and lack of trust in technology. Technology-related challenges identified by [Seethamraju and Hecimovic \(2022\)](#) also include the lack of third-party AI tool compatibility with either the existing audit support systems or client ERP systems mainly because of the resource intensive data cleansing effort required, tool customization needed to adapt to certain clients' industries, and the refinement of tools needed because the tools are still under development. On the client side, they highlight barriers such as client data quality and integration, and the potential for bias that can be difficult to detect and remove.

Organizationally, the authors document differences in resources between Big 4 and non-Big 4 firms. Big 4 firms have dedicated significant global investment in auditor upskilling in contrast with non-Big 4 firms that often rely on off-the-shelf tools and encounter the lack of IT auditors and those with skills to interpret AI tool results and face cost and training pressures. Challenges related to humans using AI tools are also identified. For example, [Fedyk et al \(2022\)](#) find that the main barrier to AI adoption is the lack of trained human capital. In addition to the need to upskill audit talent, trust in technology is found to be another drawback to AI implementation in audits because of uncertainty and the lack of transparency related to how the analyses are generated and because auditors generally do not take part in the development of the tools and therefore find it challenging to trust them ([Seethamraju and Hecimovic 2022](#); [Samiolo et al 2023](#)).

A few studies focus on the ethical implications of AI technology use in accounting and auditing. Through the lens of ethical frameworks, [Munoko et al. \(2020\)](#) perform bibliometric analysis of publications to explore practical ethical challenges that could be encountered in AI enabled audit. They address issues such as the lack of transparency, concerns related to data privacy, fairness of data use, and user autonomy and accountability ([Munoko et al 2020](#)). Furthermore, they raise questions about audit quality, professional skepticism and concerns over AI tools' influence over their users, governance of AI, and auditor competence. In another study focused on ethical AI use in accounting and auditing, [Lehner et al \(2022\)](#) perform a theory-driven literature review to identify five "challenges to AI-based ethical decision-making in accounting" (p. 119): objectivity, privacy and data protection, transparency, accountability, and trustworthiness. They highlight the need for understanding how to competently use and govern AI while being aware of its limitations.

This study engages with practitioners to build on the academic literature documenting recent emerging technology adoption in public accounting. To explore how AI is transforming auditing within large accounting firms, we ask the following research questions:

- RQ1: How is AI used in auditing?
- RQ2: To what extent is AI used in financial reporting by clients and what are current approaches to auditing client AI?
- RQ3: What are the challenges of AI use in auditing?
- RQ4: What are possible solutions to begin addressing these challenges?

3. Research method and study context

Similar to the studies documenting emerging technology implementation in accounting ([Cooper et al 2019](#); [Kokina and Blanchette 2019](#); [Kokina, Gilleran, Blanchette, and Stoddard 2021](#); and [Seethamraju and Hecimovic 2022](#)), we adopted a qualitative interview approach to the design and analysis of this study. The goal of the interviews is to provide unique, in-depth contextual knowledge about what is taking place in practice which can then equip researchers with insights for future studies ([Dodgson and Trotman 2022](#)). More specifically, we implemented a qualitative description (QD) methodology following [Eulerich, Waddoups, Wagener, and Wood \(2024\)](#), [Kim, Sefcik, and Bradway \(2017\)](#) and [Doyle, McCabe, Keogh, Brady, and McCann \(2020\)](#) which is appropriate for studies that explore a topic that has received little attention in the literature such as the use of AI in the context of audits and the associated challenges.² Characteristics of QD adopted in this study are collecting evidence from the field, using semi-structured interviews, purposive sampling to select knowledgeable interviewees, and maximum variation sampling to include participants with diverse experience who can

² The study received appropriate approval from the Institutional Review Board (IRB).

provide rich insights (Kim et al 2017).

We conducted 22 interviews with audit professionals in the largest public accounting firms whose average experience is 20.5 years, ranging from 5 years to 37 years at the time of the interview. The interviewees represented all of the Big 4 firms and one other international firm. Their roles can most generally be categorized as (1) practicing auditors (P-2; P-4; P-5; P-14; and P-18); (2) audit managing directors and partners (P-1; P-6; P-10; P-16; and P-17); (3) audit practice leaders (P-3; P-8; P-13; and P-15), and (4) technology leaders (P-7; P-9; P-11; P-12; P-19 and P-20). All interviewees were based in the United States and had knowledge and experience with emerging technology use in audits, including AI. Most had roles in their firms involving the introduction of new technologies or AI specifically into audit processes. We applied the following selection criteria for our study: each participant must have had either direct experience with emerging technologies, including AI, in audits or had experience developing audit methodology related to using AI tools, or had detailed knowledge on the subject in the context of auditing and accounting. Prior to the interviews, we created an interview guide that centered on our research questions but during the interviews we asked follow up questions tailored to each participant's unique and diverse experience and responses (Appendix A). The interviews took place between March 2022 and September 2023 via one of the videoconferencing software programs. Table 1 includes a detailed list of participants in chronological order of our interviews. We conducted repeat interviews with P-1, P-3, and P-4 to enable early participants to address the issues brought up by later participants (Dodgson and Trotman 2022).

Data analysis consisted of first condensing interview data into analyzable chunks, then assigning descriptive codes to the data based on its core content and meaning (Miles, Huberman, and Saldaña 2014; Creswell and Creswell 2018; Doyle et al 2020). Similar codes were then categorized and clustered according to the research questions and themes. The themes were not predetermined and emerged from information collected from participants; however, AI challenges were categorized using the components of the Trustworthy AI framework as a guide (Deloitte 2023a).

4. State of AI use in auditing

Audit has seen significant changes due to the development of various technologies from analytics to automation with RPA bots and automated data blending and analysis workflows, and now tools that use machine learning and AI. For an illustrative list of technologies and their categories commonly used in auditing, please refer to Table 2.

Interview participant P-1 summarizes the current composition of tools employed in audit in the following:

Right now, it's 75 % automation and 25 % analytics, visualization whereas before, it would have been 95 % analytics and visualization and 5 % automation. A lot of this though becomes integrated once we get to the end user. (P-1)

P-3 adds that the initial audit goal was to be able to perform a "data-first audit" that transitioned to implementing a "data-first audit" quickly and with high quality. Currently, the focus is on building AI tools that could guide the auditors by offering suggestions (P-3).

The process of auditing that is more reliant on technology has caused the human auditor to evolve as well by shifting the focus from

Table 1
Participant demographic data.

Participant Number	Position Title	Interview Number	Years of Experience	Company
P-1	Assurance Partner, Audit Innovation and Digital Leader	I-1	20	Company A
P-2	Assurance Senior	I-2	5	Company A
P-3	Executive Director, Global Emerging Technology Standards	I-3	23	Company A
P-4	Audit Director	I-4	12	Company B
P-5	Audit Director	I-5	12	Company B
P-6	Audit Partner, Global Assurance Technology Leader	I-6	25	Company B
P-7	Innovation Hub, Trust Technology Leader	I-6	31	Company B
P-8	Senior Manager, Audit Innovation and Data Science	I-7	15	Company C
P-9	Senior Manager, Digital and Finance Transformation	I-8	17	Company C
P-10	Audit Managing Director	I-9	25	Company C
P-11	National Managing Principal, Digital Transformation	I-10	28	Company D
P-12	Managing Director, AI Lead for Trust	I-11	34	Company B
P-13	Managing Director, Americas Professional Practice, Audit	I-12	19	Company A
P-14	Assurance Manager	I-13	5	Company A
P-15	Manager, Americas Professional Practice, Audit	I-14	7	Company A
P-16	Partner in Charge, Audit Innovation	I-15	26	Company D
P-17	Audit Partner	I-16	37	Company B
P-18	Senior Manager, Digital and Innovation Strategy, Audit	I-17	14	Company A
P-4	Audit Director	I-18	12	Company B
P-1	Assurance Partner, Audit Innovation and Digital Leader	I-19	20	Company A
P-19	Partner, US Leader of Enterprise Innovation	I-20	29	Company E
P-3	Executive Director, Global Emerging Technology Standards	I-21	23	Company A
P-20	Partner, Americas Technology Risk Leader	I-22	26	Company A

This table provides information such as participants' position title, years of experience, and company. Each participant is assigned a participant number and an interview number in chronological interview order.

Table 2

Examples of AI technologies used in auditing.

Simple AI	Common use case types	Typical use cases in audit
Rule engines Simple machine learning Computational linguistics	Robotic process automation (RPA) Predictive analytics Simple natural language processing (text analytics, speech recognition)	data-gathering across a sequence of tasks financial anomaly detection extraction of contractual terms
Complex AI/Machine Learning Deep learning neural networks Generative AI	Image recognition, predictive models Natural language processing, code generation	physical inventory counting controls reviews, code generation for data analysis

humans as the source for inquiries to data. P-17 illustrates this as follows:

What I often say is auditing is going to change from asking people questions to asking data questions; making inquiries of the data instead of making inquiries of people. So, being able to analyze data and understand relationships and knowing what questions to ask of the data is going to become a critical skill for the auditor. (P-17)

4.1. Rules engines – RPA

Similar to [Fedyk et al \(2022\)](#), we find that RPA is not widely used in audits (P-4, P-5, P-10, P-17). P-4 reports that bot development has “peaked” and that tasks or processes that could benefit from RPA automation have already been automated after a few years of working with a vendor. P-9 shares that RPA use cases in audit have not been particularly successful and that it is more efficient to use their firm’s cloud integrated audit platform to perform audit procedures. Similarly, P-10 reports that auditors are encouraged to use company-wide centralized tools as opposed to creating their own because certain standards must be met before they can be used, and regulatory inspections place great emphasis on audit tools. RPA bots are mostly used for administrative tasks such as entering travel-related data and scheduling meetings (P-4; P-5) and identifying items that have or have not been received from the client based on a previously generated request list (P-5). For audit testing based on 100 % of transactions, auditors have widely adopted an automated workflow tool (P-5). Nevertheless, there are some areas of audit where RPA bots can be found, including accounts receivable, accounts payable, and investment confirmations (P-5; P-18), and more specialized areas, such as a bot that reads shipping invoices and gets shipment information from a website (P-17). Further, P-17 adds that the firm favors investments in technology able to spot anomalies, read documents, and predict relationships. These would be candidate applications for intelligent automation or more complex AI. This observation of limited RPA use in audits contradicts with [Perdana, Lee, and Kim \(2023\)](#) who propose a more optimistic outlook on RPA use in audits. A possible reason for this more favorable RPA usage view could be that [Perdana, Lee, and Kim \(2023\)](#) develop prototyped RPA solutions for audits in the context of a five-month consulting experience and are unable to provide data on the use of their RPA prototypes beyond that limited time frame.

4.2. Machine learning (Simple AI and complex AI)

While exploring the use of automation routines in audit enabled by RPA, data blending workflows, and other similar tools is important, AI and machine learning in particular has emerged as one of the main areas that is top of mind of audit firm leaders because the technology introduces a host of new risks, dilemmas, and challenges. Accountants have commonly used machine learning for simple predictive models, but are increasingly exploring more complex forms of machine learning like deep learning neural networks and generative models. P-3 shares this concern about using complex AI in the following:

RPA is less contentious for us right now. AI and cybersecurity are the big ones that everybody’s worried about. (P-3)

P-3 highlights that some people and organizations do not consider RPA to be a form of AI, but we consistently referred to it in interviews as a simple form of AI because it makes rule-based decisions and performs tasks that could previously only be done by humans.

Unlike [Fedyk et al. \(2022\)](#) who document a widespread use of AI, we find that most complex ML tools are still being developed and piloted. For example, P-13 states the following:

The AI tools that we have today are mostly in pilot. [...] We do have some AI tools that are fully out, but that’s machine reading and OCR type stuff [...]. So, it’s pulling information from tools and stuff like that, but other ones are all in pilot. (P-13)

P-17 confirms that ML in audit is in the early stages of development:

We are playing around with some anomaly detection, looking at journal entries, but right now, it's in kind of the brainstorming stage. (P-17)

We would observe from our research that some firms are beyond brainstorming, but few have reached large-scale production deployment with these complex AI technologies.

P-7 shares that the focus of the tools is on enabling greater efficiency and that intelligence has not been implemented yet:

It's really that efficiency play versus any true intelligence. [...] It's just like faux automation, but high quality, right? It's getting us away from people having to manually tie things together. It can create connections and data that humans would never find. (P-7)

One of the tools that is currently under development at one of the firms is a platform that summarizes audit engagement metadata for multiple years to spot outlier engagements compared to their peers (P-1; P-13; P-15; P-18). The tool uses a global database of engagements to assist auditors by informing them of the scope of the engagements, identifying significant risks and significant accounts selected in those audits (P-13). Auditors are then able to benchmark their engagement against audits in similar industry, of similar company size and location and evaluate how they compare to their peers (P-13) and determine if audit strategy needs to be reassessed (P-15). P-13 provides more detail about the tool in the following:

I've identified one fraud risk. All the other companies in this sector, same size, same kind of location or geography have identified five. Well, am I missing something, or is there a legitimate reason why I only have one? The hope over time is to then say, [...] could you then start a brand-new engagement and say, this is my industry, this is my size, this is whatever and it then prepopulates based on all this learning to say, we think [...] these are your three fraud [risks] and these are your significant accounts. [...] Yeah, it would save a ton of time. [...] That's a good example I think of where AI is being tested right now. (P-13)

Another example of a tool the development of which is currently being explored in audit is "AI-guided flows" that inform the auditors of the "best next step" (P-15). However, P-15 informs that customizing the tool to the context of a particular audit will be challenging as it will likely involve having to tailor the algorithm and then be able to justify what was done to the regulators. P-15 shares the following:

One of the classic examples about AI, let's say, it's looking at a subledger and saying, well, based on my previous experience with analyzing data, you should look into these three transactions, or this account doesn't seem to belong here, there's something wrong. [...] But in accounting, sometimes it's very difficult because we're doing reasonable assurance and we're trying to control the audit risk, you don't want to issue a wrong opinion. But then you don't want to overwork, so the balance there is very difficult. [...] That ties back to the audit risk because in certain engagement you probably want to control that risk where you'd rather have a little bit more false positive and test a little bit more. That means you'll probably have to tailor that algorithm and that's where it can get really tricky and how do you defend your file but that's sort of the grand vision. We're still just at the surface of exploring. We can get the data now. Now, what we do with it and then make it really useful and then defend it in an inspection or articulate what we did, that's going to be way more challenging. (P-15)

AI is used to estimate sales and inventory levels. P-18 states the following:

We can estimate a sale at a certain place based on so many different factors like the weather, the traffic in that area, looking at all the people living within six, seven miles of that store. We can predict the inventory levels and the sales levels at certain locations based on many different factors. AI comes into place analyzing all that and tells us, these seven factors are actually not relevant and then these ten are the ones that are determining the sale level at this particular store. (P-18)

Another area where AI has found its applicability in audit is using machine vision (employing deep learning neural networks) to do inventory counts. P-7 states the following:

A machine can look at a whole warehouse in a split second or one picture and I can inventory it down to the SKU level and I can look for anomalies, find anomalies. So, full population testing [and] understanding, things that before as humans we'd sample and feel comfortable, but we don't need to do that in the new world. (P-7)

4.3. Natural language processing (Simple AI and complex AI)

Even though much of AI-related technology in audit is in brainstorming stage, tools that use natural language processing (NLP) have already been implemented. NLP can be based on computational linguistics, or more commonly in recent times, deep learning models. For example, P-7 summarizes where NLP tools are being used:

We're mostly focused on large amounts of data [...] and then using AI to get that to a point a human can understand something that he never could before, just because of the volumes: leases, contracts, financial statements, looking at public 10Ks and 10Qs and all

these things, right now reading those, but also generating those because those are pretty formulaic. [...] Next step would then be [...] generating the narrative around a financial statement. We're very close to that. (P-7)

P-6 reports that they use a tool that reads leases and summarizes common terminology in order to facilitate lease treatment determination based on the leasing standard. P-8 shares that they are focused on internal controls auditing with the assistance of an NLP-based tool that is able to accelerate the process by extracting pertinent information from documentation particularly related to complex controls that contain multiple pages and incorporate various layers of review. P-8 provides more detail in the following:

You're spending hours reading a single contract sometimes to understand. [...] What we're doing is building tools that can extract information from that documentation, not to automate the process necessarily, but to show people, look, in this paragraph, these three people's names are mentioned and these three roles. Is that right or wrong? [...] We're able to extract at a reasonably high level, something with 98 % accuracy, things like people's names and roles, the frequency of a control operating, the systems in which the control is operating, any sort of information using the control schedules, reports, documents, things that are going to be relevant to someone's understanding of the effectiveness of the design of the control that we can at least focus their attention on. (P-8)

In addition to using NLP in the audits of internal controls, an NLP tool is also used in reliably extracting information such as vendor-provided metrics from ESG reports (P-8). Furthermore, P-8 states that NLP tools are also used for extracting high-risk disclosure language from financial statements. For example, the tool can spot "the language that is indicative of a SPAC warrant" and the language that is "indicative of liability versus equity treatment" and provide that information to the user. Importantly, the tool does not make any decisions for the users, it just identifies the presence of items regardless of the appropriateness of how they were recorded. Common use of AI for document reading is also reported in [ICAEW \(2023a\)](#).

Generative AI and large language models in particular are a major future focus for large audit firms, although at the time of our interviews the technology had not been incorporated into audit processes on a production basis. However, firms are making large

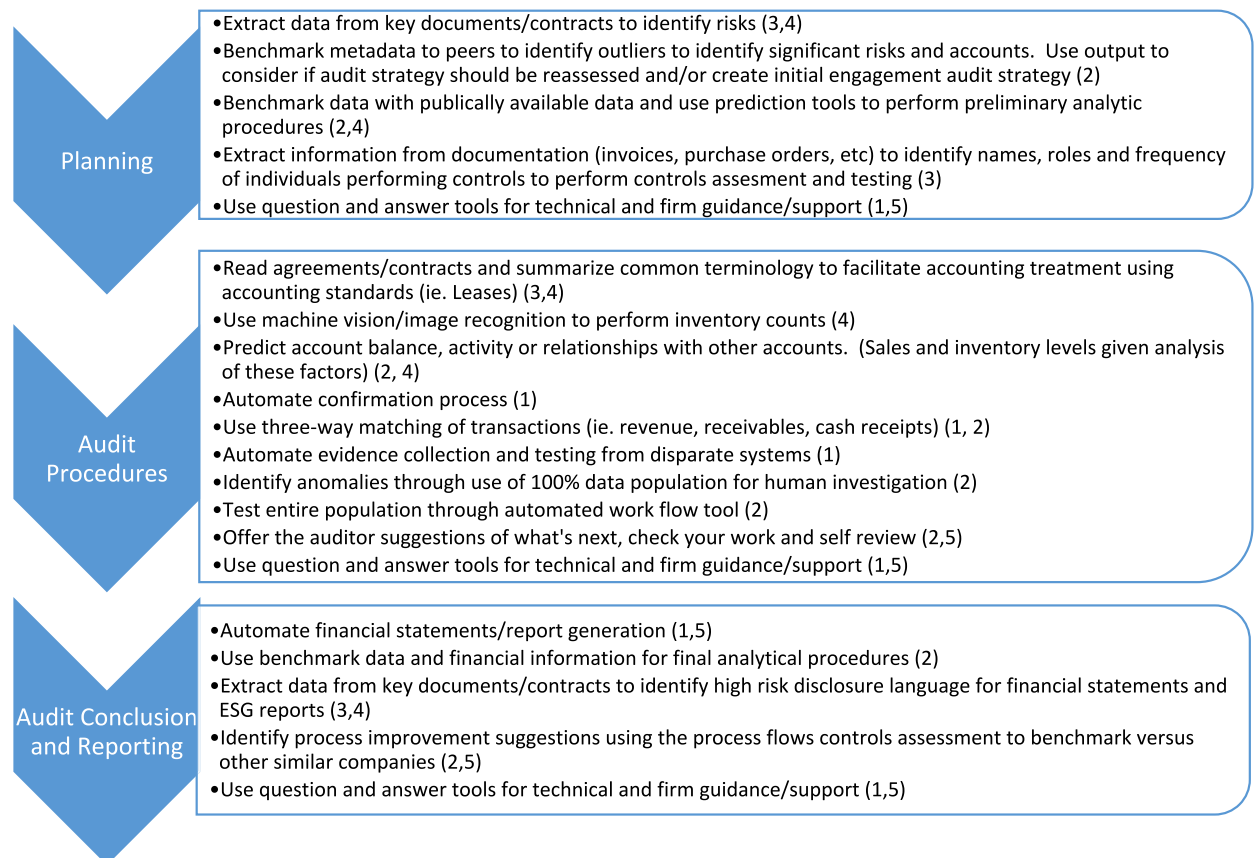


Fig. 1. Selected AI technology use (in production or experimental use) during the audit process (based upon discussion with our interviewees). Fig. 1 presents AI technology use (in production or experimental use) during the audit process. Numbers next to each technology use example refer to the type(s) of AI technologies as follows: Simple AI: (1) Rule engines, (2) Simple machine learning, and (3) Computational linguistics. Complex AI: (4) Deep learning neural networks, and (5) Generative AI.

investments in the technology and are rolling out applications for individual-level productivity. PwC's US firm, for example, has announced a \$1 billion investment in generative AI for internal use, and has rolled out a conversational system for individual use called ChatPwC. EY has announced that it is investing \$1.4 billion in the technology. Deloitte has launched an application, called "PairD", for its professionals in some regions to use for email, coding, and research (Wilkinson 2024). Although generative AI is likely to have a major impact on all audit firms, because of its uncertain outputs and occasional "hallucinations" (false information) it is likely to be only slowly incorporated into audit processes and used only with human oversight.

The technologies described are and can be used throughout the entire audit process from planning, obtaining audit evidence and completion and reporting. Overall, we find that of different AI technologies, NLP is being used in a variety of contexts in auditing while more complex ML tools are being developed and piloted. Included in Fig. 1 are examples of how simple and complex AI could be and are being used in various aspects of the audit. While both are being adopted in the audit process, more of the use cases relate to simple AI, with expectations to move to more complex AI in the future. The potential future uses of these technologies around risk assessment, internal control evaluation and substantive audit procedures offers much opportunity to improve the quality and efficiency of the audit.

5. The extent of AI use in financial reporting and approaches to auditing AI

In order to further our understanding of AI development and use in auditing, it is important to explore how audit clients are using AI in financial reporting as client use of AI often serves as a driver for AI use in the audits themselves (P-3). According to P-3, companies in industries such as high tech, banking, and insurance are more ahead than the rest with their use of AI; however, they have not yet implemented complex AI in financial reporting due to the low payoff in the space. P-3 shares the following:

What they [clients] are seeing right now is if you think about the money it takes to put in an application, and train it, and design it, and really get behind it, the payoff in a financial reporting process is not high. [...] We do not have a client yet that has a material impact on their financial statements because of AI. So, not yet in the financial reporting processes, however, we have many of them that are getting there, moving that way. (P-3)

Similarly, P-4, P-5, P-6, P-7, P-17, and P-20 confirm that audit clients are not currently using AI extensively in financial reporting. This is consistent with survey results reported by KPMG (2023). Instead, the clients are more likely to implement automation to achieve internal efficiencies. Some are exploring the use of generative AI but there are not enough solutions for financial statements yet (P-20). P-4 states the following:

Clients are using AI, almost all of them. None of them are using it for financial reporting. (P-4)

A possible reason for client hesitancy to implement AI in financial reporting is because it is a regulated environment and having significant automation and AI might expose the clients to more audit testing in accordance with IT general controls (ITGCs) unless they maintain the manual controls performed by humans. P-3 and P-5 share the following:

It's the reason why it [AI] is slow to get into the financial reporting processes because those numbers do have to be signed off by everybody and any client moving towards that needs to understand if they can't control it, it can't be audited. (P-3)
AI, they [clients] are dipping their toes in the water, I would say, but nothing that's so far impacting SOX or SOC 1 [...] because it's a regulated environment. So far, our client has said, you know, "No, we're not ready to use that technology in an area that would be in scope for SOX." (P-5)

When asked about the methodology to audit client AI, it seems that this is an area of significant ongoing discussion and that auditors are likely to audit around AI at least in the early days of implementation. P-3 discusses their approach in the following:

Something we're trying to think about with AI is, are we going to have room in the early days of AI to just ignore it? When controls and computers first took off, we audited around the computer. [...] We didn't care what the computer did because we were going to go and just beat up what came out of it. But then we moved to a controls-based approach which said, let's go into the computer and see how it's controlled and that is obviously much more efficient. We think in early days of AI, there could be an opportunity for us to go back to that mindset. We hate to do it, but given the skillset of auditors and accountants we could get around it. (P-3)

P-8 confirms that the most reasonable approach to client AI audits is to review the outputs the models generate as opposed to reviewing the code of the model itself. P-8 describes this in the context of NLP as follows:

We don't have to necessarily see what that algorithm is doing and that holds true for a lot of language models. We always got the concern initially when we are deploying this natural language processing that we deployed now in our own platform that this is going to be a black box because we are actually learning, we are using deep learning models, neural nets that are just black box... I think the argument that the engagement team uses is that, we don't need to go in and look at the code. We can just review the outputs and determine whether it's right or wrong. (P-8)

Both P-3 and P-17 emphasize the ongoing nature of methodology development for auditing client AI. P-17 shares the following:

We're using it [AI] ourselves and developing new audit techniques. So, we're developing our own guidance and then figuring out how we're going to apply that to when we audit it. [...] We're in the midst of it [developing AI guidance]. [...] unsupervised AI with no audit trail likely won't work, from an audit -- not auditable. (P-17)

In addition, P-3 describes a challenge of determining the scope of the auditors' work in cases where management is committed to controls, governance, and oversight over AI, and the challenge with determining the complexity of AI that needs to be audited:

If you think about the audit risk model and when we choose to go after control risk, we choose to take advantage of management's commitment to controls that reduces what we have to do on the detection risk. So, we're looking at that life cycle of an AI application and say, if we see how you governed and provided oversight of ideation, selection of techniques, selection of people, we are skilled enough to build this and understand it, quality data, training environment all the way through the life cycle. How much does the quality of all of that process mitigate what's left at the end and so we're trying to think through that right now. (P-3)

P-1 adds that current guides for AI audits internal to the firm are not providing much guidance. Further, P-1 provides an example of an audit of a social media company where a finding was issued by a regulator in an external review because the auditors relied on an AI/ML tool that was changing its precision levels in determining which transactions should be sent for supplemental review. As a result of the finding and because the regulators could not accept an audit team not being able to define a specific review threshold, the client had to set a new much higher threshold for transaction review in order to generate a definitive output every time even though virtually no transaction was reaching that threshold. P-1 concludes that the lack of trust in these tools discourage auditors from auditing the AI and instead continuing to audit around it.

Audit work tasks need to be auditable and have a trail. Firms seem to still be evaluating the impact of reliance on controls on these new technologies and are auditing around the technology by evaluating the outputs of the models. This is an area firms will likely be focused on to improve quality and efficiency of the audit process.

6. Firm governance structure for technological innovations in audit

An important aspect of innovation and digital transformation in audit is the governance structures developed by the firms to foster responsible use of the tools developed by auditors. Firms typically have strong governance structures in place when implementing centrally developed technology tools across practices for the sake of tool availability globally and standardization (P-1). Citizen-led automation is important for innovation and employee motivation and development. However, governance of citizen-led technology can be more challenging. P-8 states that their firm is in the process of developing governance for citizen-led automation while P-13 highlights that their firm has already developed guidance for "custom solutions." Interviewees report that citizen-led automation goes through a rigorous review and certification process before becoming available for broader use through a depository of citizen-developed solutions such as a micro-solutions marketplace or a digital lab (P-4; P-13). To use any of the tools in the depository, auditors are required to fill out a specific form in the audit work papers where they justify that the tool was fit for purpose (P-1).

When auditors design and test their own bots or workflows for a particular engagement, directors and partners are expected to provide appropriate review and supervision (P-5). The methodology for citizen-led solutions requires the design and implementation of IT general controls (P-1). Particular importance is placed on tool documentation; tools developed by individual team auditors must meet strict documentation requirements where "every step needs to be annotated" in order to be explainable to regulators (P-4). When tools are new, auditors also make sure that the automation routine they have designed is parallel-tested the "old manual way" and re-performable to achieve consistency and gain comfort that the routine is working as intended every time it is used (P-4). Finally, P-13 shares that one of the main challenges with solutions designed by audit teams and not developers is finding the right balance in establishing governance that is not too tight and restrictive because the time auditors need to meet documentation standards might prove to be too much of a burden that could ultimately hurt the rate of audit innovation. As a result of the challenges and far-reaching implications associated with using more complex AI, firms will need a centralized and updated governance process for these technologies, as discussed in the challenges of AI section of the paper.

7. Challenges of AI use in auditing

Our interview analysis indicates that there are many challenges that accounting firms encounter as they implement AI in audits. 100 % of participants mentioned trust as a key issue in AI adoption in audits. Trustworthy AI can be conceptualized as fair and impartial; robust and reliable; preserving privacy; safe and secure; responsible and accountable; and transparent and explainable (Deloitte 2023). We have organized the challenges discussed by the interviewees according to the most relevant themes and identified additional themes that are more unique to auditing such as overreliance on AI and the need for AI guidance in audits.

7.1. Transparency and explainability

The terms transparency and explainability refer to understanding the model and reasons for its outputs and is a risk associated with AI tools that use machine learning (ICAEW 2023b). Transparency of AI systems relates to the "information about datasets, processes,

uses, and outputs that are shared between stakeholders” (Ammanath 2022, 63). Explainability in AI means “it is possible to understand how an AI output was calculated” (p. 77). Moreover, explainability is related to interpretability which centers on understanding relationships in a system as well as understanding “the training, the data, and the learned components, such as weights and parameters” (p.77). Finally, intelligibility addresses “how well a model can be interpreted in human terms” (p. 78).

Challenges related to the transparency, explainability, interpretability, and intelligibility of AI tools were raised by many interviewees mainly because of the need to communicate how the tool performed to the regulators. P-13 states the following:

It's hard to train that AI, it's hard to document and support how that AI works. (P-13).

P-7 further elaborates on the concern related to model explainability in the following:

Firm standard audit tools have to go through a very different process or actually the PCAOB has to, and the AICPA may be like, we have to build these in a way that is very open and transparent to the regulators and we have to go through that process very specifically to make sure that they understand what these tools do. There's still a lot of gray area in AI and regulators are saying it's tough because AI output is pretty opaque. We don't know how it got to that answer. So, that's why we layer in the Responsible AI Framework to say, “[...] We don't know exactly how it made the decision, but it's not biased. It's following the standards and guidelines” and we've had a lot of good progress with regulatory bodies, governmental bodies to say, this is the best we can have right now, and the efficiency and the value of the AI is worth it. (P-7)

Some are developing approaches to improve model explainability (Zhang, Cho, Vasarhelyi 2022), although this capability requires considerable data science expertise. For example, P-8 suggests:

I Can dig a little bit deeper and say, “What's this thing actually doing?” Let's not dive into the code just yet. What can we do to make this more interpretable to us and there's a variety of different ways you can do that. Different techniques that are out there, LIME, Shapley, if you've heard of these different things that try to get to explainability. [...] different forms of regression to get to feature importance. (P-8)

Others are using the methodology developed for clients' use of AI, which centers on clients' ability to explain their AI to the auditors, as the blueprint for implementing AI in audits. P-3 shares the following:

How do you provide the trust in AI that everybody needs to say the information I am getting is correct. [...] You can't give somebody information that they make a decision on and it's not the right information, but it's the same question that our clients are having. If they are putting artificial intelligence into their financial reporting processes, their controls can't get them there. They need to have a way of telling us how they know it's right and so we're working on methodology for the clients' use of AI, but it becomes a mirror image. It's going to be applied to our own AI tools. (P-3)

Communicating how AI develops its outputs will be a challenging area for firms to work through with their clients and regulators. While different techniques are starting to be used to explain this information, lack of knowledge and the complexity of these models may hinder some from relying on the technology for financial reporting areas. As a result, in cases where machine output is not sufficiently transparent, auditors might default to judgments based on their existing knowledge of the client (Samiole 2023).

7.2. Fairness and impartiality

Related to challenges of transparency and explainability is a concern over AI bias and thus fairness and impartiality. AI bias refers to the phenomenon of an algorithm producing results that are systemically prejudiced (e.g., by favoring one group of outcomes over another) due to erroneous assumptions in the training process. AI, if not sufficiently controlled, can then spread the bias found in the data used in its training (Deloitte 2023). P-17 shares the concern over AI bias in the following:

The one thing personally I worry about with AI is data bias. How do we make sure we're feeding it really a representative data population, so it's not giving us a biased answer because I gave it biased data. (P-17)

Of note, AI bias in auditing is not necessarily related to demographic or socioeconomic characteristics of individuals like it is in other fields, but rather it refers to incorporating sufficient information in data sets and proper training of algorithms to adhere to the correct regulations. P-20 shares:

We are training a system, and we are inputting the regulations (i.e., US standards, international standards, PCAOB). Perhaps we only upload three of four regulations so as a result, the model is biased against the fourth regulation [...] When we train –did we upload the right information—did we include the right regulations, did we include a complete data set? (P-20)

To address algorithmic bias, the firms are developing tools to audit AI. For example, P-7 states:

We are building tools to audit AI. We have been engaged, maybe not for audit clients, but to audit the bias of AI models. (P-7)

However, there are challenges with using AI tools for detecting AI bias in auditing, although such tools are being developed in computer science and used in other fields (Lee, Resnick, and Barton, 2019; Roselli, Matthews, and Talagala, 2019). P-20 states:

The PCAOB standard on auditing says we can't assume or indirectly test something. We need to have direct testing of a control. Inferring it works (through the machine learning algorithm) versus having to directly test the control – will the PCAOB adjust their rules for this? (P-20)

Because much of the AI bias stems from the underlying data, several interviewees discuss a few data-related challenges. One of those roadblocks is accessing data from client ERP systems. P-7 states:

It's too risky right now and again the systems that are working across these ERP systems are pretty clunky as we know it, because we have to get the data out of them. AI needs good data to be effective and we're just not there yet. (P-7)

P-20 suggests focusing on data architecture because if data is not architected consistently and with proper cybersecurity, then errors will be introduced into AI.

Furthermore, P-12 believes that using client data alone might not be enough as it is unlikely that it will contain the needed degree of outliers and variability. One potential solution is to supplement real client data with synthetic data but ways of implementing this is an ongoing challenge:

Even if you have access to hundreds of client data sets, you may still not see all the things you need to see to prove your tool is robust enough to be able to detect everything that the regulatory bodies expect to do. [...] In the future the remaining challenge is how we can augment the real-world data with potentially synthetic data [...]. (P-12)

The profession has some significant ongoing challenges related to AI bias, both actual and perceived. While no clear solutions have been identified, firms will need to find ways to address finding representative data populations that are supported by good quality data.

7.3. Privacy

Privacy in AI refers to “guarding sensitive information, gaining consent for use of that data, ensuring models are resilient and do not leak or divulge protected data, using models in a way that respects privacy, and meeting the emerging laws and regulations around the world that do not just encourage but indeed mandate privacy” (Ammanath 2022, 128).

Data privacy becomes particularly important at the AI tool training stage. P-3 discusses a concern related to the ethical use of client data in accordance with data privacy considerations when training a tool based on a vast number of contracts such that a client could become identifiable:

Another issue that we're struggling with is [...] that we would never monetize our client's information. Let's say you were building a document reader that pulls out all of the lease terms of a huge amount of contracts. If you train it on one client's data, it's going to learn some things and then it's going to go apply that to another client's data and all of a sudden – a knowledgeable auditor could read between the lines and start seeing the other client – understanding the other client's data. [...] General counsel will not yet say we're in a position to learn across clients. (P-3)

P-18 states that clients are concerned about providing the auditors access to vast quantities of their data in fear of privacy and being audited around the clock. P-18 shares the following:

Even now the clients are hesitant to extract data, provide data, because there's always a hesitancy, okay, “Well we don't want to give you all our data, are you guys going to be auditing us 24/7 now?” [...] The sizes of the data is still a problem, the security of the data is still a problem. [...] We get the entire inventory movement for the entire year and then we can just mine that, so we can tell you how you're recording your revenue from start to finish and when you're recording it, when you're collecting the cash. We can already do some of these things. If we have more data, we can do more. (P-18)

To address client data privacy concerns, P-20 suggests only using publicly available client and industry data to build models. P-8 describes another approach to using client data to train their AI tool while ensuring their privacy. Specifically, P-8 suggests that even though they are not permitted to use client data to develop benchmarks for clients in the same industry, they are able to extract certain pieces of client data such as control language and train their AI tool on the different ways that information can be presented. The tool uses that learning only to adjust the weights of the algorithm. P-8 outlines their approach to using client data as follows:

It took us a long time to get everybody comfortable with the concept of using client data to build, train and produce models. [...] Some of our clients overtly expressly prohibit us from using their data and we abide by that, full stop. [...] The way that we train and utilize client data is in a way such that we're not permitted to disclose client information. We're permitted to use it by the professional standards, but we're not permitted to disclose it. [...] By no means can [we] benchmark a client against somebody else. I would love to be able to say, “[...] We have a client in oil and gas within your equivalent of order to cash cycle. We've identified 10

individuals in your process flow that have a certain role in that process where your peers average three people, which means that perhaps you're a little bit more highly manual than your peers, maybe you could use an ERP upgrade, some automation, some different back-end connection upgrades." We can't do that. [...] "Let's keep this to things that are going to help serve more universal means, like information extraction, like being able to extract those entities from that control language that I mentioned, that we use data from a variety of different clients intentionally, so that we can train it on different ways that different information can be stated." That's how we get the power of it, but we're not necessarily regurgitating anything that the client provided to us. It's really just being used to alter the weights of the algorithm. (P-8)

Producing strong AI models requires large volumes of data to train the models. The more data, the more accurate results the model will produce. It is critical for firms to protect client data and build trust with clients. It will be important for firms to be transparent around how the data will and won't be used and demonstrate strong governance in this area to convince clients to provide access to more data.

7.4. Robustness and reliability

An AI model is likely to have an issue with robustness and reliability when a model performs well in training but fails to operate well in a dynamic environment after it is deployed (Ammanath 2022). Several interviewees addressed a concern regarding the ability of auditors to rely on tools that are changing after deployment. For example, P-3 shares this challenge in the following:

I Can tell you that the AI in it is very concerning to us. When we put a tool out, we go through what we call global certification. That's the quality of the fit for purpose, the functioning, the data that was used, the methodology that goes with it. It's very hard to certify something if it changes the next day or a week later. We've been very hesitant about how quickly we can move with AI. (P-3)

P-13 highlights the ongoing debate that surrounds the issue of allowing the tool to learn from real data once the AI tool is in production where it is potentially becoming more robust and reliable while considering the lack of ways to actually measure the accuracy of that tool. P-13 states the following:

It's tough from a technology person's perspective because that's where the real learning happens. You can feed it as much dummy data or training data as you want, but in production you're then increasing the volume of data 100X, right? That's where the real benefit would come. We have those internal debates all the time. In fact, I just had one the other day of a tool that wants to use AI and push it to practice to get that feedback from the practice to say, oh, this recommendation was right and this one was wrong. It's like, well, I don't know if we can do that yet because we don't have a mechanism yet to measure the accuracy or effectiveness very well because we don't know what it's doing yet. It's a challenge for sure, especially in our regulated environment. (P-13)

Even though the use of continuous dynamic learning might be considered in early AI adoption discussions, we predict that audit firms will not use it due to the regulated nature of the industry. Furthermore, in rare cases where dynamic learning could be tested or used, there would be ways to measure the accuracy of predictions from machine learning models using MLOps systems (Kreuzberger, Kühl, and Hirschl 2023).

According to P-8, there is a way for auditors to control the evolving nature of AI tools due to dynamic learning. P-8 informs that they allow a model to be retrained only during specific time periods that typically coincide with various platform updates. This approach prevents exposing auditors to the risk of a model reaching different conclusions at different times during an audit and the risk of a model drift which occurs when a model becomes less accurate over time. P-8 states the following:

The way we've deployed our tools, they don't keep learning. They are, by design, not dynamically learning in a reinforcement learning loop. The way we've addressed it is by having a manual retraining process. We'll look, we'll assess to determine if there's any drift in the way that our models are working based on the data that's being applied. Again, a good example is information extraction routine. Is it still extracting, generally pretty reliably based on what people are inputting? But we don't retrain it really midstream because we don't want to create this paradigm of having two classes of audits during an audit year or even just a single engagement having a switch that they're not aware of. And if they were to run the model at the end of the audit, they would reach a different conclusion. [...] So we have fixed time periods in which we're applying updates that align with the major updates of our platform in general so that it's not just a surprise, we're going to put this out there. [...] Right now, nothing is dynamically learning. (P-8)

Although challenges exist associated with the dynamic learning aspect of the technology, it appears as though relying on training during specified periods may allow for more consistency and stability across clients. While there may be some sacrifice of improving the model with more recent data during these periods, consideration of the reliability, consistency and the ability to test the model may be more important.

7.5. Auditor overreliance on AI

The challenges discussed above focus on the technological side of AI tool design and deployment. While these challenges are

significant, it is equally important to address another theme that emerged from interview data analysis – challenges that the human auditors are likely to encounter when using AI tools. On the one hand, some interviewees (P-8, P-13) share that it will take time for auditors to gain comfort with using AI tools. On the other hand, interviewees are also concerned with potential auditor overreliance on AI tools (P-7, P-13).

P-13 suggests that in order for auditors and regulators to familiarize themselves with AI tools and test them, the tools will be released in “baby steps” using AI-generated clusters by industry sector or sub-sector. AI will be running in the background and auditors will not directly interact with AI (P-13). P-8 adds that when the early iterations of an AI model were deployed, auditors were assured that the model only provided suggestions that they could activate if they wanted to but were not required. The auditors were encouraged to use the tool to double check their work and for self-review, but it was not a formal audit procedure (P-8).

As familiarity with the tools grows among auditors, the firms are recognizing the human tendency to over rely on technology and are developing approaches to prevent auditors from doing so. For example, P-13 shares that the tool that evaluates lease contracts for non-standard terms requires the auditors to verify its accuracy by providing the screenshots of those contracts and asking auditors to check that the data were selected correctly. This is done because the firm cannot make a case for relying on the tool 100 %. P-7 informs that auditors are required to exercise their professional judgment in determining the extent to which they can rely on AI. P-7 states the following:

For any of our tools, we have required either training or guides and or in-app guidance that tells the auditor in not so many words: this is the degree to which you can rely on this and this is what you need to do supplementary to that to be comfortable that your work is true and you can rely on your work. So, we don't have any tools yet where we just say, “Oh, poof, here's the answer auditor, you're done” – that doesn't exist. (P-7)

P-13 calls for continued exercise of professional skepticism and critical thinking by auditors when interpreting the output of AI tools. For example, if an AI tool suggests that an auditor's decisions are the same as the peers', an auditor might wrongfully rely on this information when in fact something in that engagement makes it different from its peers. P-20 suggests focusing on educating auditors to understand where the data is coming from and how the data flows from beginning of the process to the end.

As with many technologies, concerns around overreliance on the tools exist for AI as well. Audit approaches are being adjusted to incorporate additional procedures to reduce the likelihood of this happening. In addition, auditors need to understand the tools are making suggestions, however the auditor should still continue to use their professional judgement and skepticism when using these tools. Furthermore, greater emphasis should be placed on designing audit support systems that minimize the deskilling that affects accounting professionals (Arnold, Collier, Leech, and Rose 2023).

7.6. Trust in client AI-generated reporting

Among the common cases where genAI is used or expected to be used are writing disclosures of financial statements and writing code for various reports (CAQ 2024). Some of the risks reported are completeness, accuracy, and appropriateness of these disclosures as well as model performance (CAQ 2024). Echoing concerns expressed in the CAQ report, P-20 mentions that code generated by ChatGPT could be used by clients to make a change in the system that would introduce a vulnerability if a person using the code lacks programming competence. Currently, some of the risks can be mitigated by placing a human in the loop for manual checks and reviews. In the future, however, genAI-related risks are likely to become more difficult to assess as reliance on such technology increases.

In the context of auditor–client communication, some interviewees discussed the dilemma of trusting information provided by clients if genAI becomes more common in generating responses to auditor requests, especially in the remote work environment. For example, P-20 states:

How do I know that what it came up with is the right stuff? How do you know the response you got wasn't generated by ChatGPT?

In light of the risks discussed, determining the source of the information may be challenging resulting in more difficulty for auditors to trust the information provided by clients. Additional skepticism may be required.

7.7. Need for regulator guidance

Some interviewees express concern over the lack of frameworks that govern simple automations which in turn makes them question the readiness to govern AI especially because of the difficulty with transferring human thought process and judgment to AI. P-1 states the following:

You don't have a framework that even governs simple automation, you definitely are not ready [...] as I bring in an AI engine that's going to drive decisions that affect the control framework. [...] I'm going to process it to bring in all this other qualitative factors where it's now taking over somebody's judgment that we already don't do a good job of being able to translate into, how did I apply that judgment, what was my judgment thought process. We're nowhere near ready to let the machines do that and cover that in a risk framework. (P-1)

In addition to not having a governance framework for automations, challenges that seem to resonate with several interviewees are

the pressure of regulatory scrutiny (P-13) and the absence of an AI auditing standard (P-3). P-3 shares the following:

The European Union right now is actually focused on AI. Again, everybody begins with governance because you cannot just do this haphazardly [...] but nobody's actually written a standard on it yet. They're all starting to focus on it. So, the IASB, which is your AICPA or PCAOB, has a working group focused on technology. How are we using it? How are clients using it? What does it mean? Everybody is recognizing the need for definitive guidance in all of these areas. (P-3)

P-17 calls for the need to modernize the standards in the U.S. and points out that "there's nothing in the standards that prevents you from doing it [using AI], but it also doesn't help you do it or encourage you to do it". P-18 adds that "regulation is not catching up with the developments". P-18 further explains that they are informing the PCAOB of their use of AI in audits by providing them with use cases to demonstrate that "hey, regulation needs to catch up [...]; by enabling you to run, you guys are actually developing the next generation auditor." (P-18).

8. Addressing the challenges of AI adoption: Summary of lessons from our interviews and other fields

The challenges with adoption of AI in auditing listed in the previous section are most generally related to technological limitations, human factors, and audit tasks / phases; however, the boundaries between these categories are not clear-cut. Table 3 summarizes our main findings on challenges with AI adoption from Section 7, pointing out explicitly the main categories to which they belong (technological, human, or audit task). Solutions proposed by the interviewees are also listed.

We note that challenges with AI adoption are not unique to the auditing field. Other fields, such as healthcare, face similarly complex issues. For example, although there is huge potential in utilizing AI for real-time patient monitoring, ambient control and wellness, and privacy compliance, a number of obstacles exist because of healthcare operational procedures, verifiability of prediction models, resilience, and lack of ethical and regulatory frameworks (Saraswat et al 2022). Learning from other industries and from developments in computer science may be helpful in identifying additional solutions the audit profession can use to tackle the challenges of implementing AI into the audit process.

8.1. Addressing technological limitations

The main technological challenges identified from our interviews – transparency and explainability, fairness and impartiality, privacy, and robustness – are an active area of research in computer science that falls into the realm of Responsible AI. The explainability of AI is a key characteristic that can be used for AI diagnosis and analysis to ensure that objectives of transparency, ethics, and accountability are met.

Transparency and Explainability. It is widely recognized that AI systems need to construct explanations for their output to improve system understanding and behavior predictability, calibrate system trust, as well as support system debugging, verification, and justification (Hayes and Shah, 2017; Chakraborti et al., 2019; Mercado et al., 2016; Kulesza et al. 2015; Biran and Cotton, 2017). Explainable AI (XAI) is a growing field with numerous important players from both academia and industry, including Microsoft with Azure ML Workbench, Google Cloud AI, Kyndi with its XAI platform for government, financial services, and healthcare, and FICO with its Credit Risk Models (Adadi and Berrada, 2018). From our interviews, it was clear that many auditing firms already employ explainability techniques (Table 3).

The different purposes of explainability and transparency in AI models may result in profoundly different requirements for explanations (Lipton, 2018), and there are substantial difficulties with categorizing and evaluating developed methods for XAI (van der Waa et al, 2021). Explanations can vary by *style* (the way the information is structured, often algorithmically generated) and *form* (how the information is presented, e.g., textually or visually).

The style of explanations can most generally be categorized as *rule-based* and *example-based* (van der Waa et al, 2021). Rule-based explanations are based on "if...then" rules, e.g., in the auditing context, one might state that if accounts receivable days outstanding increased from last year and are higher than the industry average, then the system would recommend you perform certain additional audit procedures. If accounts receivable days outstanding do not increase from last year and are not higher than the industry average, then no additional procedures are recommended by the system. Some foundational ML models, such as decision trees and regression, are inherently rule-explainable because they easily demonstrate the effect of various features (inputs) on model recommendations. With more complex models, one can utilize feature attribution approaches such as Local Interpretable Model-agnostic Explanations (LIME; Ribeiro, Singh and Guestrin, 2016) and approaches that build on the concept of Shapley values in economics (Shapley 1953; Rozemberczki et al. 2022; Sundararajan and Najmi, 2020), to determine the marginal contribution of a feature to model predictions. More recent methods unifying concepts from both such as SHAP (Lundberg and Lee, 2017; Molnar, 2023) have been gaining in popularity. Metamodels have been used in healthcare applications, where simpler inherently interpretable models are trained on the outcomes of complex AI models to explain how the complex models make their recommendations (Zhong, 2020; Zhong et al. 2022; Canellas et al., 2023). In general, when building complex models, it is a good idea to include features that those who need to receive explanations inherently understand and then maintain traceability of those features through the model.

Example-based explanations may refer to historical explanations in which the advice was found to be true or false. In the auditing context, using the example above relating to accounts receivable, system example-based explanations may provide the auditor output that recommends additional audit procedures, but it would also include examples of other situations from the past (i.e., prior year data for the same client or current year data from another client in the same industry) that would allow the user to understand the system

Table 3
Challenges and potential solutions of AI-aided audit support system use in audits.

AI audit support system challenge category	AI audit support system challenges addressed by interviewees	Potential solutions proposed by interviewees	Challenge category		
			Technology	Human	Audit tasks / phases
Transparency and Explainability	How to communicate and explain AI output to the auditors and to the regulators in order to have trust in the accuracy of AI-generated information.	- Employ different explainability techniques: LIME, Shapley. - Use different forms of regression to get to feature importance. - Apply the methodology that is being developed for client use of AI by the consulting arm of the firm.	X	X	
Fairness and Impartiality	- How to make sure that the data used in training AI is representative of the population and the outcomes are free from algorithmic bias. - How to access data from various client ERP systems. - How to determine if synthetic data needs to be generated because available client data is insufficient to train AI tools.	- Firms are building tools to audit the bias of AI models. - Focus on ensuring consistent data architecture with proper cybersecurity.	X		
Privacy	- How AI tools can be trained in an ethical manner ensuring client data privacy and security. - How AI tools can be trained without enabling the tool to learn across clients. - How can client fear of being audited around the clock caused by enabling auditor access to vast amounts of client business data be addressed. - How can AI tools use client data only for improved algorithm performance while ensuring clients cannot be identified, disclosed, or benchmarked against.	- Extract certain pieces of client data such as control language and train the AI tool on the different ways the information can be presented so the tool uses that learning only to adjust the weights of the algorithm and not for benchmarking which is not permitted because of data privacy considerations. - Use publicly available client and industry data to build models.	X	X	X
Robustness and Reliability	- How auditors can rely on a tool that is learning post deployment and might reach different conclusions at different times during the audit. - How firms can certify tools that are changing for use in audits globally. - How the accuracy and effectiveness of the AI tool can be measured. - What are best practices for assessing model drift which occurs when a model becomes less accurate over time.	- Control the evolving nature of AI tools and prevent model drift by allowing a model to be retrained only during specific time periods that coincide with platform updates. - Disable dynamic learning based on auditor feedback and don't retrain AI mid-audit.	X		X
Auditor adoption of AI	- What are the best practices for auditors to adopt on AI tools. - What are best practices for auditors to not over rely on AI tools.	- Release AI tools gradually by industry sector or sub-sector. - Assure auditors that AI model only provides suggestions for self-check and its use is not a requirement nor a formal audit procedure. - Require the auditors to verify the accuracy of the AI tool by enabling them to check that the data were selected correctly. - Require the auditors to exercise their professional judgment, professional skepticism, and critical thinking in determining the extent to which they can rely on AI. - Focus on educating auditors to understand where the data is coming from and how the data flows from beginning of the process to the end.		X	
Need for AI Guidance	- How definitive AI guidance can be designed in the absence of a framework that governs simple automations (e.g., RPA) or the use of AI that is more complex. - How to successfully transfer auditor thought process to AI. - How to codify auditor decision-making process.	Inform the PCAOB of AI use by supplying them with AI use cases in audits.			X

Table 3 presents a summary of challenges and potential solutions related to AI adoption in auditing provided by study participants.

decision in comparison to these other situations. Google Cloud AI, for example, utilizes an “approximate nearest neighbor service” to return to the user similar examples to new predictions or instances (<https://cloud.google.com/explainable-ai>). When the recipient of the explanation is human, example-based explanations can be more effective than rule-based explanations (van der Waa et al, 2021).

Fairness and Impartiality. Conventional fairness techniques include importance weighting, data transformation, and variable blinding to correct for data curation protocols, as well as the elimination of specific features that should not be ethically included (Diao et al, 2021; Chen et al., 2023). Algorithmically, incorporating measures of fairness and impartiality in AI models can be done by actively adjusting model parameters during training. Commonly used metrics include demographic parity, decision theory cost, equalized odds, equal opportunity, and equal quality of service (Lundberg, 2023). However, it should be recognized that the definition of fairness inherently depends the context in which machine learning models are used, and it is important to define measures of fairness carefully in AI algorithms so as not to obscure important value judgement choices. Some popular explainable AI tools like SHAP allow for decomposing measures of fairness and allocating responsibility for any observed disparity among each of the model’s input features. As pointed out in Lundberg (2023), “Explaining these quantitative fairness metrics can reduce the tendency to rely on them as opaque standards of fairness, and instead promote their informed use as tools for understanding how model behavior differs between groups”.

Inadequately selected model specifications can introduce bias even when other measures (such as properly vetted data sets) are taken to reduce such bias. For example, Obermeyer et al. (2019) show an example of how label choice in predicting medical care costs can work against Black patients. The bias derives from a predictive tool’s reliance on claims data – information contained in patients’ medical bills – rather than on data contained in patients’ clinical records about their health. In auditing, thoughtful selection of input features that matter for measured outcomes is critical for unbiased use of AI.

It is important to note also that once AI models are deployed, predictions start affecting outcomes, and perpetuate existing system biases. (See Pachamanova, Tilson and Dwyer-Matzky (2022) for an example from healthcare.) Considerations for fairness should not be limited only to algorithm design but also to other aspects of auditing operations, including input data, model parameters, visualizations of outcomes, and embedding of model results in the auditing workflow. (See Ahmad et al. (2020) for specific solutions from the healthcare field, and Liu et al. (2018) for a general discussion of the issues.)

Privacy, Robustness, and Reliability. The more data AI trains on, the more robust its predictions. However, the quest for better accuracy is in inherent conflict with client privacy. Our interviewees all pointed out the importance of client privacy in building AI tools in-house.

The healthcare industry has been a leader in the governance around AI and data privacy, although issues still persist (Ebeling, 2021). Some basic rules that can strengthen privacy protection include consideration of user consent, de-coupling data from users via anonymization and aggregation; enforcing strict control over who in the company has access to specific datasets; continuous audit of how the data are utilized; and testing algorithms to find the minimal amount of data that result in viable applications. Our interviewees suggested several other approaches to think about client privacy specific to auditing (Table 3).

The robustness and reliability of AI solutions in auditing is contingent on the interoperability of various solutions, which is not to be taken for granted. Each AI application typically requires a separate tool, and the tools do not necessarily work well together. We are observing through our research that audit firms focus on developing their own tools. On the one hand, this addresses many of the technological challenges. On the other hand, it inevitably slows down the rate of adoption of AI in audit.

8.2. Addressing human and audit process challenges

Technological challenges have a human aspect and therefore important consequences for the adoption of AI in auditing. Our research has unveiled that solutions typically need to rely on both human vigilance and audit process changes.

Transparency and Explainability and Auditor Adoption of AI. In some industries such as healthcare, for example, the lack of explainability in fields like radiology (which uses complex deep learning-based image analysis) has led some clinicians to ignore AI results when they conflict with the clinician’s own judgment (Lebovitz, Lifshitz-Assaf, and Levina, 2022). Extensive ongoing model and sensitivity analyses are necessary to test an AI-aided system and generate sufficient trust that the outcomes are consistent with correct practices. Our interviewees suggested that human checks be installed and that auditors be ensured that AI models provide only suggestions and their use is not a requirement or a formal audit procedure (Table 3). Rigorous and regular employee training that emphasizes both the generation of recommendations and the utilization of recommendations should be part of auditing practices. Regular employee training has also the potential to address other human challenges to adoption pointed out in Samiolo et al. (2023), including the potential for individual deskilling effects of AI technology adoption in auditing and the risk of the auditor abdicating its responsibility for judgment (Munoko et al., 2020). Instilling healthy skepticism towards model recommendations and ongoing human employee training can help trained human reviewers recognize instances in which the AI-aided system does not perform as expected. On the other hand, engaging employees in reporting such instances and establishing mechanisms for incorporating employee feedback into the AI-aided system can not only improve the system itself, but also increase trust and participation among employees.

Privacy, Robustness, and Reliability. Data privacy for client data is, of course, an important issue for maintenance of trust in audit firms. Our interviews suggested that auditors are beginning to use aggregated client data to train machine learning models, but they hastened to say that data had been anonymized before being aggregated and analyzed. Large firms that are pursuing generative AI have created private models so that prompts involving client situations or any other topic are not used to train publicly-available

models. This is a factor in other industries as well; healthcare organizations are perhaps the most restricted in terms of privacy of patient data. Mayo Clinic, for example, is making extensive use of AI, but it requires that models be trained on deidentified patient data (Lindquist, 2022).

Maintaining machine model robustness and reliability over time is another challenge that necessitates human oversight. Models can “drift” when the data on which they were trained no longer represents current reality. Some organizations are beginning to explore continuous learning and training approaches that would automatically retrain models if drift is detected. Even though these approaches are being explored in other contexts, such as marketing and banking, to the best of our knowledge production models are not dynamic as humans are still responsible for deciding when to recalibrate the models. So far there have been academic efforts to approach this, but not in practice, and certainly not in the audit industry. Audit firms, however, are beginning to use machine learning operations (known as MLOps) systems. These systems are repositories of metadata describing all models that are in production and are an important aid to managing model assets and risks. They also can monitor the prediction accuracy of models to detect drift. Use by audit firms of MLOps models is in its early stages, however (Falticeanu, Li, and Maliutin, 2022).

There are substantial regulatory barriers to some uses of AI in other industries. In health care, the FDA must approve any “Artificial Intelligence/Machine Learning-Based Software as a Medical Device” before it can be used in clinical settings. In banking, the Comptroller of the Currency regulates the use of AI and is sensitive to issues such as bias and lack of transparency in credit approval models. To date, there are no such regulatory barriers in auditing: the PCAOB is only considering proposals for AI-related regulation. Informing the PCAOB of AI use by supplying them with AI use cases in audits can help codify the auditor decision-making process when it comes to AI use.

9. Discussion and conclusions

9.1. Summary of findings

While the development and deployment of AI in external audit processes is underway, it is still in the early stages. Automation and data extraction tools—the more established AI applications—are widely used. RPA is mostly used to perform administrative tasks while its use in audits is not as widespread. More complex machine learning models that summarize audit engagement metadata to identify anomalies, AI-guided flows that suggest next audit steps, tools that estimate certain financial statement items like sales and inventory levels are either piloted or just beginning to be applied. Tools that use NLP have received wider adoption and are used in reading and generating financial statements, extracting key terminology from lease contracts, internal controls documentation, and ESG reports. Most of the auditors we interviewed, however, fully expect that these AI tools will be widely used in the future, and that they will have a substantial impact on audit firms and the activities of auditors.

Regarding client use of AI, we find that the return on investment for AI in financial reporting is limited which explains why clients hesitate to implement it in addition to further safeguards that need to be designed in order to use AI in a regulated space. The development of the methodology for how to audit client AI is underway, and auditors mostly plan to audit around AI as opposed to through AI due to its complexity, difficulty with determining the scope of audit work, and the lack of AI guidance from the regulators.

Our research indicates that the most prevalent challenges with the adoption of AI in auditing are transparency and explainability, AI bias, data privacy, robustness and reliability, fear of auditor overreliance on AI, and the need for AI guidance. A novel concern that was revealed in the interviews, and perhaps the most complicated AI-related concern to address, is the potential use of generative AI tools to structure client communication to the auditor that could be challenging to detect. Auditors have no control over their clients’ use of AI, but need to audit around AI used in financial reports that might not be trustworthy. Although this is not an immediate concern given our findings on low use of AI among clients of audit firms we interviewed, it will become a concern in the future. The introduction of regulation that sets expectations for the use of AI in financial reporting and auditing can begin to address this concern.

Some participants report a few solutions they have employed to address some of the main challenges associated with AI implementation. For example, for improved transparency and explainability, techniques like LIME and SHAP can be used. To address AI fairness, the firms are building tools to audit the bias of AI models. To ensure privacy of client data, participants report either using publicly available client and industry data to build models or only using client data to adjust the weights of the algorithm and not for benchmarking. For robustness and reliability, some report dealing with model drift by allowing the model to be retrained only during certain periods of time and disabling dynamic learning functionality. To prevent auditors from over relying on AI tools, participants report requiring auditor verification of the work of AI and encouraging the use of the tools but only as suggestions. Finally, participants share that they inform the regulator of their AI development and implementation strategies. We note that identification of potential solutions has not been fully validated in the auditing field; it is an ongoing process given AI’s rapid advancement across multiple fields.

9.2. Building data, technology, and human infrastructure

Employing AI capabilities successfully in accounting and auditing requires considerable investment in data, technical, and human infrastructure (Davenport and Mittal 2023). Large accounting firms are already making such investments. Perhaps the most important of them is to create a client data platform, usually based in the cloud for easier data transformation and access. External audits are based on a company’s core transaction data, which is often in different formats across clients. Several accounting firms have built data platforms that gather client data, integrate it across organizations, and make it available for analysis. Some firms are also partnering with external vendors to create such data capabilities.

The availability of integrated data across clients will eventually make possible benchmarking across organizations, and prediction

of financial and operational problems using ML and training data from multiple companies. Thus far, however, these value-added capabilities using client data are not yet in broad use or are only experimental.

To make extensive use of AI, accounting firms need to build technology infrastructures that make it possible. They will need to acquire a variety of AI software tools, develop partnerships with AI software and hardware vendors, and either move to cloud-based data storage and processing, or build up substantial computing resources on their own premises.

Perhaps most important are the human capabilities necessary to build and manage AI systems in accounting firms. Firms are already hiring data scientists, but a few such professionals will be unlikely to be sufficient for broad-scale usage of complex AI. In addition to more data scientists, firms will need to hire or train many “citizen data scientists” on how to create, analyze, and interpret client data and predictions about their businesses. They will primarily use simple AI technologies, developing automation tools and simple models for their own personal work and sometimes for enterprise-level tasks. In addition, as professional services firms, they will need many experts who can help clients with their AI needs in accounting and finance, operations, and other aspects of their businesses. Both Deloitte and PwC, for example, have created programs to train as many employees as possible in basic AI understanding and citizen data science skills. PwC has also trained citizens in automation programs (Barkin and Davenport, 2023).

9.3. Future outlook

Wider adoption of AI systems offers opportunities for accounting firms to dramatically change auditing processes and the role of human auditors. It seems likely that increasing numbers of audit tasks will be automated, and that machine learning models will identify likely anomalies and problems with financial reports. The need for further investigation of financial transactions will be prioritized by data and models. Human auditors in this scenario will be expected to perform detailed investigations, review the work of machines, and sign off on the integrity of AI-enhanced audits. It is too early to say whether this will lead to less need overall for human auditors, but that is certainly a possibility. One new study of the impact of generative AI systems on different job types suggests that up to half of human accounting tasks could be performed by this type of AI (Elondou et al., 2023). Thus far, however, the primary role of AI relative to human workers has been augmentation rather than automation-based job loss. Increasing numbers of jobs involve working alongside AI, which can offer the benefits of both increased productivity from AI and human flexibility and innovation in the work process (Davenport and Miller 2022). In the context of auditing, Samiolo et al (2023) describe how AI/ML tools enhance or “scale up” an auditor’s capability by performing tasks that a human auditor is unable to perform. We expect that augmentation will be the dominant approach in accounting and auditing for at least the next decade or so.

Accounting firms may also develop additional client service activities in the area of algorithm and AI assurance (Deloitte 2023b). The firms do not now audit the ethics of AI systems, because (at least in the US) there is no government-sponsored standard with which organizations are expected to comply. However, this situation may change in the future, and there are some companies that provide auditing and assurance for AI systems. For example, Holistic AI, a London-based company, has created a process and platform to manage the process of reviewing AI assurance for several large clients. The platform allows clients to assess all proposed AI use cases, and to specify various details about the system—the purpose, business case, the project’s ownership, team composition, the data used, the type of AI technology employed, whether it is being developed internally or by an external vendor, degree of autonomy, and so forth. The platform uses the answers to such questions to score the application in terms of its potential risk. The risk domains include explainability, robustness, efficacy, bias, and privacy. An increasing percentage of the evaluations in the Holistic AI platform are based on the European Union’s AI Act (The AI Act, 2023), which also ranks AI use cases in three categories. The act will take effect in 2024. Even though the act applies only to European businesses, some multinational companies plan to adopt it globally, as has been the case with the EU’s GDPR data privacy legislation.

9.4. Limitations and directions for future research

Our study has a few limitations that offer directions for future research. This paper captures the state of AI development and current challenges with the adoption of AI in auditing – areas that are continuously evolving. P-20 stressed the need to learn from academic research about the risks related to AI implementation and mistakes made in auditing and other fields to assist auditors when they encounter AI in audits and when they develop their own AI. Future research could revisit our research questions to document more current AI use cases and investigate whether AI-related challenges we discussed persist in the future and, if not, what solutions were developed to manage the challenges. For example, studies could explore approaches to AI system training and the data used in the process as well as ways to deal with continual learning of AI tools and their explainability. Studies could focus on documenting commonalities and differences in perspectives based on the different roles and tenure levels of professionals within the firms. For example, we observed that both audit partners and auditors were more focused on the day-to-day use and reliance on AI tools while audit practice leaders and technology leaders were more concerned with various aspects of responsible AI and auditor overreliance on the tools. The technology leaders, as one would expect, were focused on the use of AI tools in the long-term and strategically identifying other areas where these tools could be used. Future studies could also investigate the ethical considerations associated with AI tools and methodologies and frameworks developed to audit the ethics of AI algorithms. Furthermore, studies could research the applications and impact of generative AI in auditing in light of recent developments in the field (Financial Times 2023, Wall Street Journal, 2023).

Studies could also examine the benefits of AI use in audits. Some benefits mentioned by our participants were increased standardization of audit procedures across clients (P-13), better audit quality due to fewer human errors (P-6; P-7), and improved ability to assess cloud-based tool usage and measure efficiency gains (P-7). Future studies could research how smaller accounting firms approach

using AI in their audit practice because the interviewees whose AI-related experiences and perspectives are reflected in our study represent only large accounting firms. Finally, studies could explore the impact of AI tools on auditor roles and skillsets.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

Interview Questions.

Research Question	Key Interview Questions	Exemplary Follow-up Questions
RQ1: How is AI used in auditing?	- How is AI being used in audits? - How widespread is AI use in audits? - What are some areas where AI is being developed for the use in audits? - Can you provide some examples of AI use in audits?	- What is the timeline for development of these AI tools? - Are there some AI tools that are widely used while some are in the pilot stages? - Is AI that is used intelligent or is it mostly automation?
RQ2: To what extent is AI used in financial reporting by clients and what are current approaches to auditing client AI?	- To what extent are your clients using AI in financial reporting? - What are the approaches currently used to audit client AI?	- How are you approaching developing tools to audit your client AI? - What are some of the issues you have encountered or could encounter when auditing client AI?
RQ3: What are the challenges of AI use in auditing?	- What are the challenges of AI use in auditing? - As you develop AI tools for your auditors, what additional considerations do you need to keep in mind?	- How will the AI systems be trained? - How is AI output explained? - What kind of data is used to train the tools? - How will auditors deal with the continual learning of AI? - What are the ethical considerations associated with these tools? - Describe what your company means by an AI Responsible Framework?
RQ4: What are possible solutions to begin addressing these challenges?	Can you share how you have or are addressing any of the AI challenges?	What potential solutions to AI challenges will you consider addressing in the future?

Appendix A displays key interview questions and examples of follow-up questions we asked during the interviews.

Data availability

The data that has been used is confidential.

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